Study III-D
Precision-based confirmation of cross-session information

Registered design development and independent simulation audit

XAU/USD | Asian realised variance -> subsequent London realised variance | 11 October 2026

Decision

The frozen precision/validation/capacity gates do not authorise fresh confirmation.

The prespecified minimum-score selection returns **None within the frozen grid**. Existing OANDA timestamp coverage permits at most **1,975** fresh scored dates after reserving 400 training dates. This is an optimistic ceiling, not a quality-audited empirical sample.

No pre-2016 numerical prices were opened, no fresh forecast outcomes were evaluated, no strategy was run, and the final 2024+ holdout remains closed. Study III-D is a new design and does not amend III-C.

What changed

The scientific question is unchanged. The decision framework now requires a positive dependence-robust interval and a predefined maximum uncertainty, rather than a 90%-power requirement against a 5% effect. Changing the framework does not by itself create more precise evidence.

All results below are synthetic planning evidence using only already-viewed 2016-2023 calibration. Synthetic observations and 252-step subperiods are never presented as historical market dates.

1. Question, estimand and precision

Does the Asian state contain reproducible incremental information for London RV beyond HAR? Define Delta as HAR QLIKE loss minus HAR+Asia QLIKE loss. A positive mean favours HAR+Asia.

The operational estimand is the long-run expected loss advantage of the particular frozen pair fitted to the first 400 eligible fresh labels, conditional on that fitted pair and a sufficiently stable confirmation-period distribution. It is not an unconditional retraining effect or a causal intervention in the gold market.

A fixed scale prevents denominator ambiguity

The already-viewed pilot fixes **S0=0.149185243**, the mean HAR QLIKE after training on 2016-02-01 to 2017-09-01 (400 eligible rows) and forecasting 2017-09-05 to 2023-12-29 (1,504 rows). This is nuisance calibration, not fresh confirmation.

The maximum two-sided full interval width is **0.10 x S0 = 0.014918524** raw QLIKE units. Thus the half-width resolution is about +/-5% of the fixed pilot baseline loss. It is not a 10-percentage-point CI for fresh relative improvement; the fresh denominator would be different and random.

The realised fresh relative gain, mean Delta / mean fresh HAR loss, is a secondary descriptive statistic. It does not replace the frozen precision denominator.

Prospective choice	Frozen rule
Minimum considered scores	1,000; at least four populated annual subperiods
Planning grid	1,000 / 1,250 / 1,500 / 1,750 / 1,975 / 2,500 / 3,000 / 4,000
Precision assurance	95% MC lower bound >=80% for meeting width in every scenario/signal cell
Evaluation schedule	All eligible fresh data to 2015-12-31; one evaluation; no favourable stopping

2. Measurement and causal forecasting

Window	Clock and role
Asia	09:00-15:00 Asia/Tokyo = 00:00-06:00 UTC; current feature
London	08:00-12:00 Europe/London; DST via named timezone; subsequent target
Daily	UTC00:00-24:00; only completed prior daily RV is a predictor

Use native unsmoothed M5 midpoint candles. One common set of squared-return edges is partitioned into A, L and remainder R; D=A+L+R. Consecutive received closes must be five minutes apart within a UTC day. The first daily bar uses open-to-close. Gaps are never bridged or filled.

HAR uses previous valid log L, previous valid log D, and log prior 5/20-valid-L means. HAR+Asia adds current log A. Previous single-session/daily values must be no older than four calendar days. All rolling windows exclude the current London target.

Both log-RV models fit intercept and unpenalised OLS on exactly 400 initial eligible labels. Back-transform using the training-only mean exp(residual), then freeze coefficients and smearing. Subsequent available history updates predictors only. Variance predictions have a 1e-12 floor; nonfinite values stop the procedure.

Quality and issuance

Retain III-B quality criteria: sessions >=95% of expected M5 bars, both boundaries, max received gap10min; daily >=85%, max gap90min. Flag unordered/nonfinite OHLC, nonpositive open/close bid-ask spreads, spreads above USD5, and adjacent-bar opening jumps above2%. No flagged session is rescued by deleting individual bars.

At06:00UTC only completed Asian and prior-window quality can determine issuance. Current London or later-day quality cannot cancel an already issued forecast. A dedicated timing test checks this; another compares the entire reconstructed 2016-2023 feature ledger with the frozen III-C pilot.

Source-specific confirmation remains OANDA-only. A new period on the same provider is fresh chronology, not independent-feed replication.

3. Full-procedure simulation and risk reference

The III-C generator is reused without rewriting its frozen files. It generates positive Asia, London and remainder RV recursively and constructs daily RV exactly as their sum. Every future HAR input comes from generated history. Forecast fits remain ordinary OLS even though the nuisance generator has its documented stability constraints.

Scenario family	Innovation resampling
Core dependence	Circular residual blocks5,10,20
Persistence stress	Blocks40
Volatility stress	Blocks20; all log-innovation residuals multiplied by1.25
Signals	Zero channel; existing 1%,5%,10% long-training calibration targets

Asia innovations are resampled independently of joint London/remainder innovations. Within-stream dependence and London/remainder joint dependence survive; the full original joint distribution does not. The intervention controls the Asia channel rather than shifting forecast losses.

Each independent synthetic path uses 500 burn-in steps, 400 training observations, and up to4,000 scored observations. Every fitted pair has an independently integrated stationary risk reference: 64 paths x8,192 scored steps, with separate seeds and burn-in. The reference uses the same prediction floor, fitted coefficients and training-only smearing.

Reference uncertainty is measured across independent paths. Calibration and coverage use a +/-2.58 integration-SE bracket; independent half-reference coverage and oracle-SE/evaluation-SE are also reported. This is approximate integration of a stationary risk, not an exact finite-horizon conditional expectation.

Completed: 8,000 calibration fits, 20,000 independent validation fits and 2,560 reference trajectories. The generator stopped on any containment/nonfinite failure; recorded component violations: 0.

4. Interval construction and error calibration

For each loss path, compute the exact conditional circular-bootstrap variance of its mean at blocks5,10,20, including the short final block. Take the largest SE. No inner-bootstrap Monte Carlo noise enters this SE.

The primary interval is **mean Delta +/- 3.063399 x maxSE**. The common critical value is calibrated from the highest 97.5th percentile of absolute studentised reference error plus integration uncertainty, over all planning N/scenario/signal cells, with a minimum1.959964. It was saved before independent validation.

This deliberately conservative, simulation-calibrated interval has a block-bootstrap SE; it is not a percentile bootstrap interval and does not have a distribution-free95% guarantee. Calibration across the finite planning grid helps protect prospective N selection within the stated scenarios.

Independent validation at the known OANDA ceiling

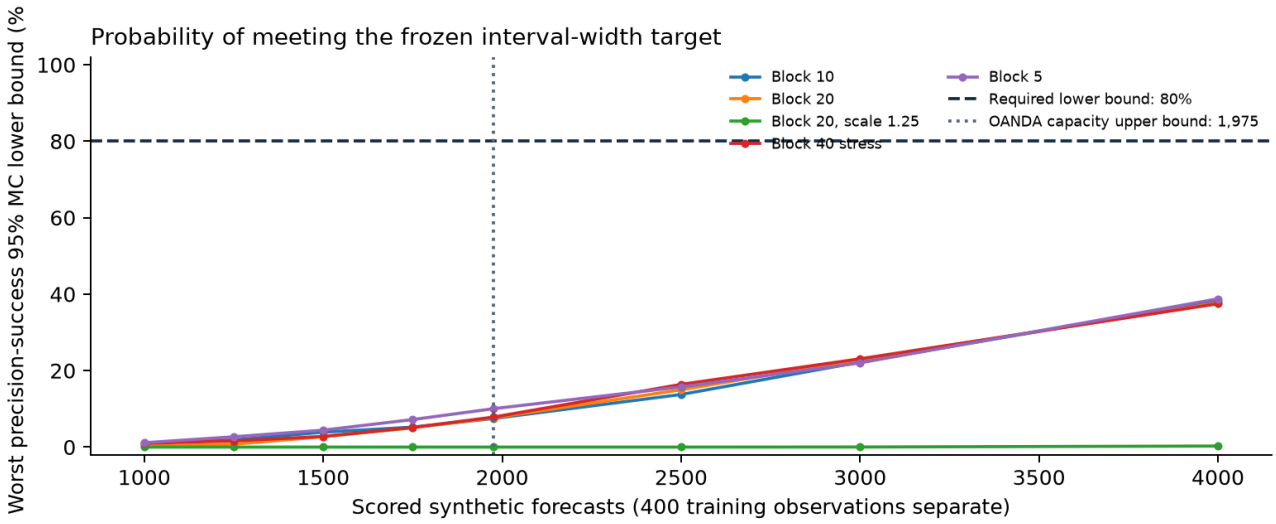
Scenario	Coverage lower*	Boundary upper*	Null upper*
Block 10	97.9%	0.7%	0.4%
Block 20	97.1%	0.6%	0.4%
Block 20, scale 1.25	96.2%	1.2%	1.0%
Block 40 stress	98.6%	0.7%	0.6%
Block 5	98.4%	0.4%	0.3%

* Monte Carlo bounds: two-sided95% Wilson lower for full reference-bracket coverage; one-sided95% Wilson upper for upper-tail noncoverage and zero-channel false positive lower-bound decisions. Worst signal is shown except the zero-channel column. Gates: coverage lower>=93%; boundary/null upper<=6%.

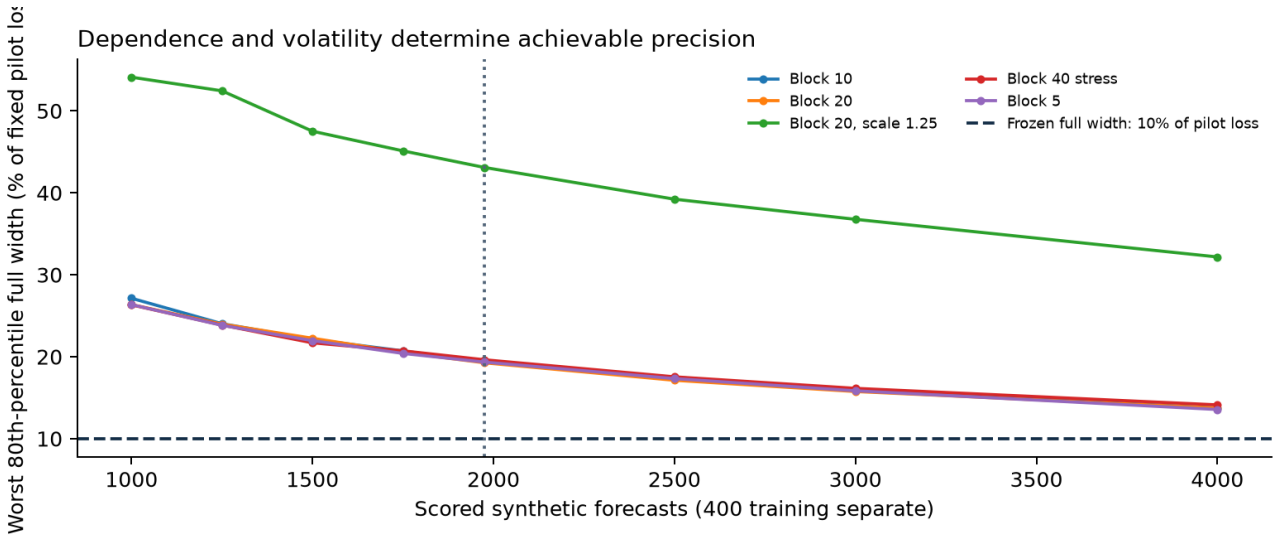
A zero-direct-Asia-channel model is not an exact equal-forecast-risk null after finite estimation. Consequently its rejection rate is an operational false-confirmation diagnostic. Boundary noncoverage and full reference-risk coverage assess interval error separately; neither is a universal size theorem.

The worst oracle-SE/evaluation-SE 95th percentile at N4,000 is 0.272; frozen limit0.20. Maximum difference between the two half-reference coverage estimates: 0.9%.

5. Precision sufficiency



Each curve takes the worst signal at that scenario/N. The plotted quantity is the two-sided95% Wilson lower bound for the proportion of independent paths whose full interval width is <=10% of fixed pilot loss. All curves must clear80%; significance probability is not the selection criterion.



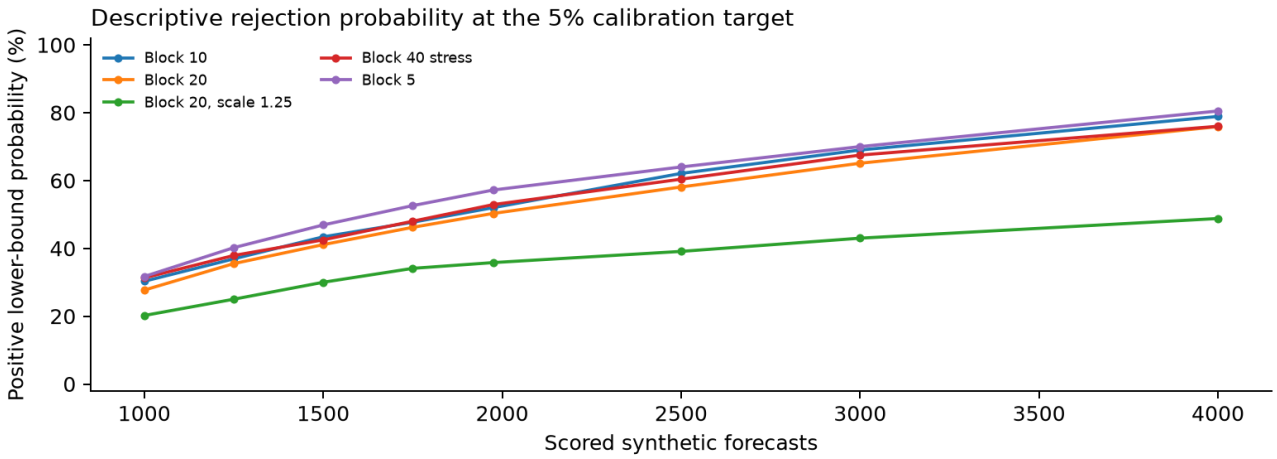
The second chart shows the worst 80th-percentile width across signals. Its10% line is the unchanged precision target. The vertical line is the1,975-score coverage ceiling, before price-quality exclusions. The cells are synthetic and do not estimate an observed pre-2016 effect.

6. Frozen sample selection and descriptive power

N	Coverage	Boundary	Precision	All gates
1,000	Pass	Pass	Fail	Fail
1,250	Pass	Pass	Fail	Fail
1,500	Pass	Pass	Fail	Fail
1,750	Pass	Pass	Fail	Fail
1,975	Pass	Pass	Fail	Fail
2,500	Pass	Pass	Fail	Fail
3,000	Pass	Pass	Fail	Fail
4,000	Pass	Pass	Fail	Fail

Null and oracle gates are included in "All gates"; full values are in SAMPLE_SIZE_GATES.csv. Select the smallest N whose gates pass at that N and every larger listed N. No alternate threshold or narrower scenario set is selected after validation.

Selected minimum: None within the frozen grid. This rule uses prospective simulations only. If no N passes the finite grid, it does not imply that no larger design could ever pass; it does stop this frozen candidate.



These curves show only the probability of a positive lower bound at the5% long-training calibration target. Achieved finite-400-training gains and the stricter joint acceptance rates are recorded separately. They are not power guarantees or empirical confirmations.

7. Source capacity and empirical decision

Requirement/evidence	Status
III-C required scored dates	4,000; unchanged
Existing OANDA ceiling	1,975, after400 initial training labels; coverage-only
III-D selected minimum	None within the frozen grid
III-D capacity eligibility	Fail
Fresh numerical prices opened in III-D	No
Fresh forecast confirmation run	No
2024+ final holdout	Unopened
Trading translation or execution	Not run
Route A	Administratively open for provider evidence

The existing audit found complete native-M5 candles beginning19March2006 and ending31December2015. A1999 search bound is not a claim of1999 history. Coverage, positive-RV and quote-quality exclusions can reduce the1,975 upper bound; nothing in III-D manufactures extra dates.

The III-C power/capacity failure remains unchanged. III-D changes the prospective objective and uncertainty requirement, not the underlying unique-date count, and not the substantive forecasting question.

Current action

Freeze the prospective failure; retain quarantine and holdout closed.

Provider answers may still establish a qualifying historical source. They require instrument-specific timestamp evidence, original two-sided quote provenance, historical clocks, source-change documentation and suitable licence rights. A different source does not automatically inherit OANDA measurement equivalence or these simulation results.

Olsen and Tickdatamarket enquiries were sent6October2026. LSEG acknowledged a sales contact request; its form accepted an abbreviated topic and needs the full checklist on follow-up. No provider suitability finding is inferred from sending an enquiry. No additional outreach is made in this phase.

8. Prospective confirmation specification

This specification is inert unless design validation and capacity pass and a public source-specific freeze authorises opening quarantine. The cached-only runner has no candle-API acquisition code and checks design, registry, code hashes and predecessor preservation before accessing numerical fresh inputs.

Gate	Predefined rule
Sample and evaluation	All eligible issued/scorable dates after the first400 labels through2015-12-31; N>=registered minimum; >=95% issued-label coverage
Incremental QLIKE	Calibrated primary lower bound>0
Precision	Full interval width<=0.10 x frozen pilot HAR loss
Asia slope	Frozen fitted coefficient>0
MSE/MAE	Each enhanced/baseline ratio<=1.05
Calendar consistency	>=4 years with>=100 scores; positive yearly paired mean in>=ceil(2/3) of those years
Sensitivity	Report all years; calendar-position paired percentile bootstrap9,999 draws at blocks5/10/20
Further holdout	Separate final protocol;2024+ stays closed here

Primary blocks count consecutive scorable forecasts, not elapsed calendar days. The secondary percentile procedure retains missing/unissued business-day positions. Simulations model complete eligible-step sequences; arbitrary calendar missingness and outcome-dependent quality exclusions are not calibrated generator features.

If source quality reduces eligible N below the minimum, stop before model fitting/loss evaluation. A sample or precision failure is reported honestly; no threshold relaxation, selective year omission, model refitting or repeated confidence-interval checking is allowed.

Public GitHub versioning provides a prospective, reviewable record before numerical fresh access. It is not an OSF registration, institutional endorsement or an independently certified timestamp. This development report does not claim a passed empirical test.

9. Evidence limits, references and preservation

Interpretation limits

The coherent generator is conditional on the viewed pilot law, stability constraints and intervention in residual dependence. Block40 and1.25-scale scenarios stress two dimensions; they do not cover arbitrary structural breaks, biased RV, old-provider microstructure, all missingness mechanisms or every heavy-tailed law. Reference integration is approximate and targets stationary conditional risk.

M5 equivalence within OANDA establishes agreement with the stated measurement construction, not that measured RV is an unbiased latent-variance proxy. QLIKE robustness results require assumptions; MAE remains only a secondary practical guardrail. HAR is adapted across sessions, not an exact replication of the original daily HAR-RV model.

Primary references

Fulvio Corsi (2009). A Simple Approximate Long-Memory Model of Realized Volatility. HAR motivation only; our cross-session HAR includes a prior-day variance control and is not an exact replication of Corsi.

Andrew J. Patton (2011). Volatility forecast comparison using imperfect volatility proxies. QLIKE proxy-robust rankings require a conditionally unbiased proxy. Native M5 equivalence does not prove that assumption. MAE is a secondary guardrail, not a proxy-robust variance ranking guarantee.

Raffaella Giacomini and Halbert White (2006). Tests of Conditional Predictive Ability. Motivation for forecast-method evaluation and estimation uncertainty. Their rolling finite-memory framework is not claimed as a theorem for our one-time frozen fit.

Xiaofeng Shao and Dimitris N. Politis (2012). Fixed-b Subsampling and Block Bootstrap: Improved Confidence Sets Based on P-value Calibration. Motivation to evaluate block-size-dependent finite-sample coverage, not a claim that our calibrated max-SE interval implements their specific method.

Robert Lunde and Cosma Rohilla Shalizi (2017). Bootstrapping Generalization Error Bounds for Time Series. Risk inference needs mixing/smoothness assumptions. Our generator validation is conditional evidence, not a proof that all those assumptions hold for XAUUSD.

Audit trail and immutable predecessors

Verified 283 tracked predecessor files unchanged. Source-lock hashes cover the reused III-C routines and pilot derivatives; per-path calibration/validation statistics and all160 validation cells accompany this report. Original Studies I/II, Study III Design Pack, III-B, III-C and Route A remain intact.

Development scope and implementation were committed before new simulations. Calibration critical values were committed before independent validation. Final registration records which gates passed or failed, code/protocol hashes and continued outcome closure. Protocol choices and unsuccessful design outcomes are retained rather than overwritten.

No completion percentage or profitable-strategy claim is made. The report is a prospective methodology result, not an investment recommendation.